DM – The inclusion–exclusion principle

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Outline

1. The inclusion–exclusion principle
   - Two sets
   - Three sets
   - General case

2. Applications
   - An alternative form
   - The number of surjective functions
   - Derangements

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The union of two sets: Example

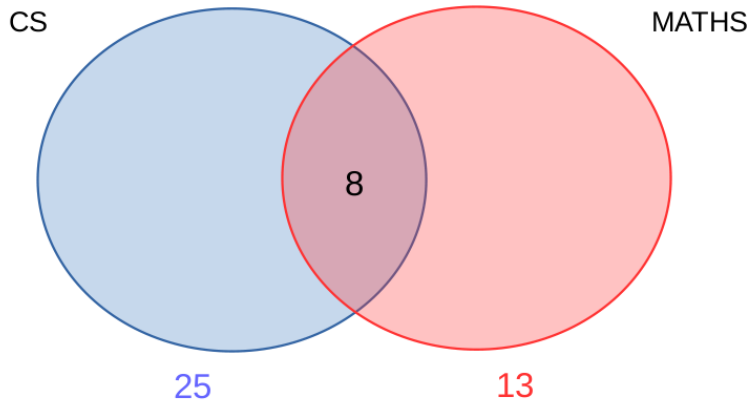
In a discrete mathematics class every student is a major in computer science or mathematics, or both. The number of students having computer science as a major (possibly along with mathematics) is 25; the number of students having mathematics as a major (possibly along with computer science) is 13; and the number of students majoring in both computer science and mathematics is 8. How many students are in this class?

Let:

- A be the set of CS majors,
- B be the set of maths majors.

We have:

$$|A| = 25\,, |B| = 13\,, |A \cap B| = 8\,.$$

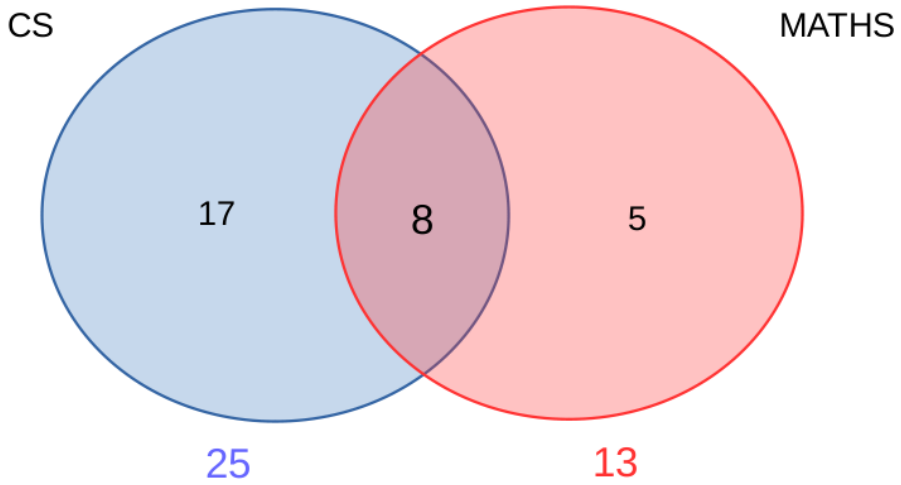


Question:

$$|A \cup B| = ?$$

The union of two sets: Example

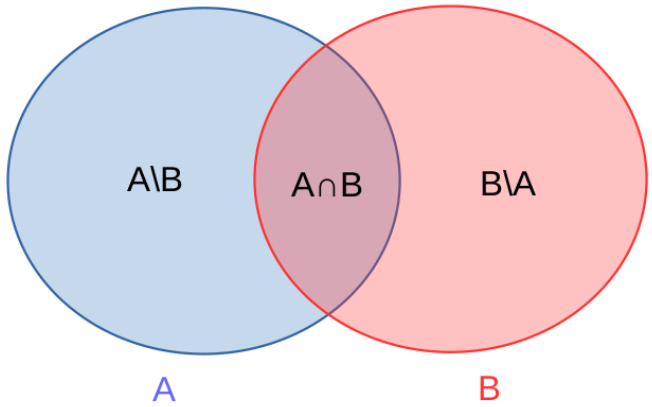
$$|A| = 25\,, |B| = 13\,, |A \cap B| = 8\,.$$



$$|A \cup B| = 17 + 8 + 5 = 30\,.$$

The union of two sets

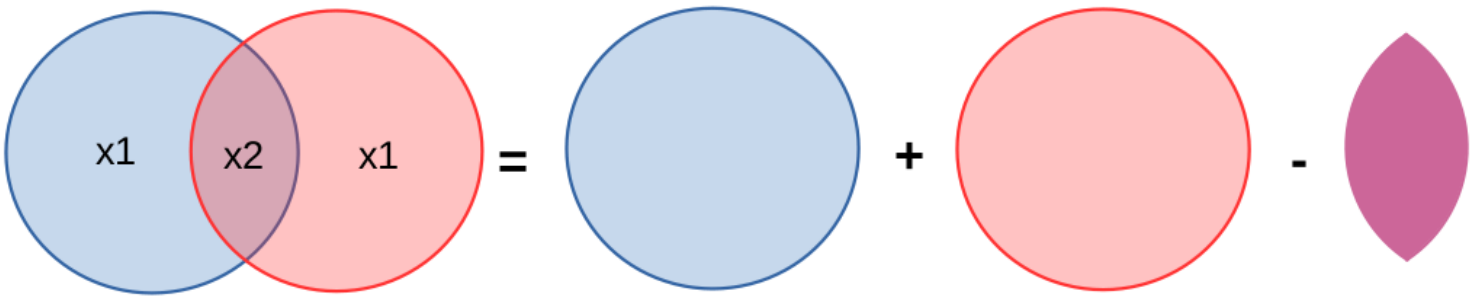
$$|A \cup B| = |A| + |B| - |A \cap B| .$$



$$\begin{aligned} |A \cup B| &= |A \setminus B| + |B \setminus A| + |A \cap B| \\ &= (|A| - |A \cap B|) + (|B| - |B \cap A|) + |A \cap B| \\ &= |A| + |B| - |A \cap B| . \end{aligned}$$

The union of two sets

$$|A \cup B| = |A| + |B| - |A \cap B| .$$



Double counting argument:

- In $|A \cup B|$, each element is counted exactly once.
- In the sum $|A| + |B|$, each element of $A \cap B$ is counted twice.
- Therefore, one count of each such element must be discounted from the RHS, which explains the term $-|A \cap B|$.

The union of two sets: Example

How many positive integers not exceeding 1000 are divisible by 7 or 11?

Let:

- A be the set of positive integers not exceeding 1000 which are divisible by 7,
- B be the set of positive integers not exceeding 1000 which are divisible by 11.

We have:

- $A = \{7, 14, \ldots, 994\}$, $|A| = \lfloor \frac{1000}{7} \rfloor = 142$;
- $B = \{11, 22, \ldots, 990\}$, $|B| = \lfloor \frac{1000}{11} \rfloor = 90$;
- $A \cap B = \{77, 154, \ldots, 924\}$, $|A \cap B| = \lfloor \frac{1000}{7 \cdot 11} \rfloor = 12$.

Hence

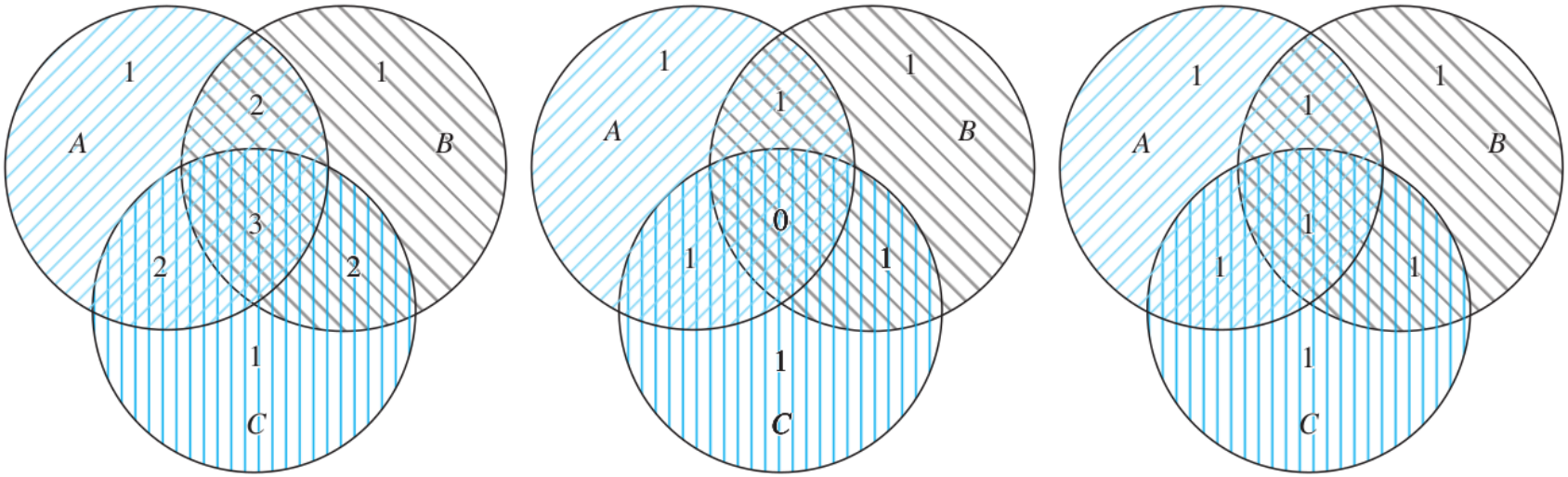
$$|A \cup B| = |A| + |B| - |A \cap B| = 220\,.$$

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The union of three sets

$$\begin{aligned}|A \cup B \cup C| = &|A| + |B| + |C| \\ &- |A \cap B| - |B \cap C| - |C \cap A| \\ &+ |A \cap B \cap C|\end{aligned}$$



(a) Count of elements by $|A|+|B|+|C|$

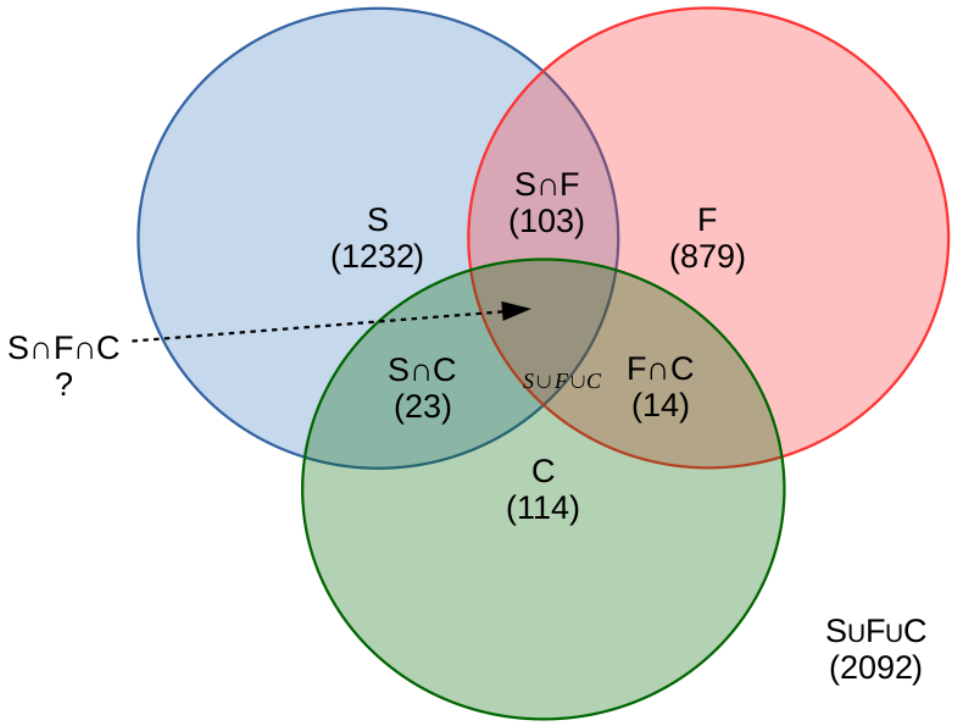
(b) Count of elements by $|A|+|B|+|C|-|A\cap B|-|A\cap C|-|B\cap C|$

(c) Count of elements by $|A|+|B|+|C|-|A\cap B|-|A\cap C|-|B\cap C|+|A\cap B\cap C|$

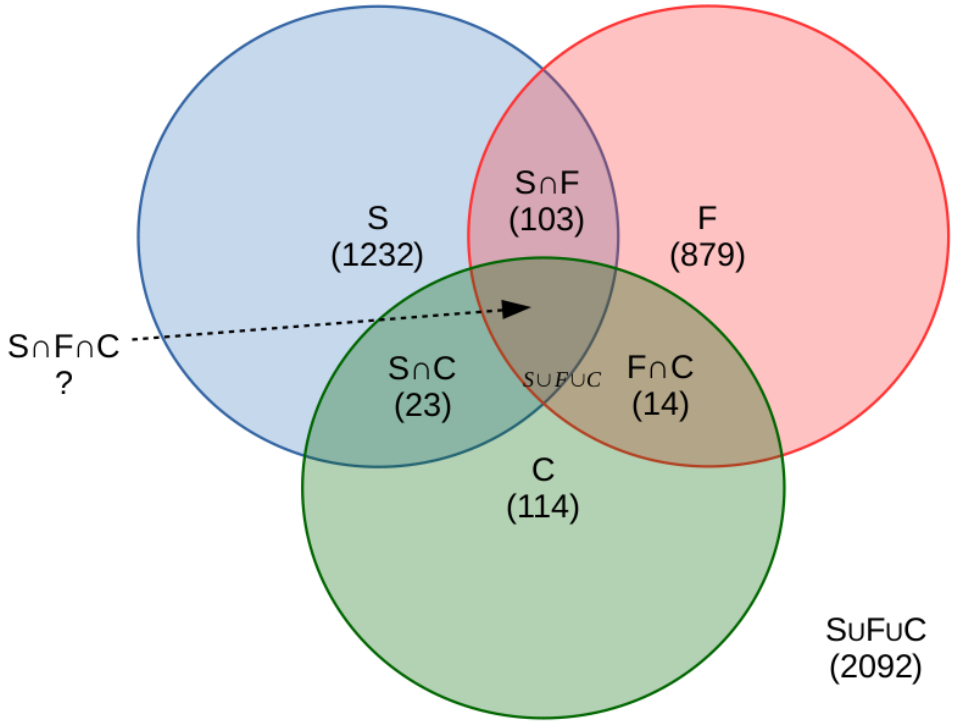
Figure: Rosen, p. 555

The union of three sets: Example

A total of 1232 students have taken a course in Spanish, 879 have taken a course in French, and 114 have taken a course in Chinese. Further, 103 have taken courses in both Spanish and French, 23 have taken courses in both Spanish and Chinese, and 14 have taken courses in both French and Chinese. If 2092 students have taken at least one of Spanish, French, and Chinese, how many students have taken a course in all three languages?



The union of three sets: Example



$$
\begin{aligned}
|S \cap F \cap C| &= |S \cup F \cup C| - (|S| + |F| + |C| - |S \cap F| - |F \cap C| - |C \cap S|) \\
&= 2092 - (1232 + 879 + 114 - 103 - 14 - 23) \\
&= 7 \,.
\end{aligned}
$$

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The inclusion–exclusion principle

Theorem

Let $A_1, A_2, \ldots, A_n$ *be* finite *sets. Then*

$$
\begin{aligned}
|A_1 \cup A_2 \cup \ldots A_n| &= \sum_{1 \le i \le n} |A_i| - \sum_{1 \le i < j \le n} |A_i \cap A_j| + \sum_{1 \le i < j < k \le n} |A_i \cap A_j \cap A_k| - \ldots \\
&\quad \cdots + (-1)^{n+1} |A_1 \cap A_2 \cap \cdots \cap A_n| \\
&= \sum_{s=1}^{n} \left[(-1)^{s+1} \sum_{1 \le i_1 < \cdots < i_s \le n} |A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_s}| \right] .
\end{aligned}
$$

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Alternative form

Consider a finite set of N elements and n properties $P_1, P_2, \ldots, P_n$. We want to count the number of elements having none of those properties.

Let P_i' denote the negation of the property P_i. Let A_i denote the set of elements having the property P_i and let

$$N(P_{i_1} P_{i_2} \ldots P_{i_s}) = \left|A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_s}\right|.$$

We have:

$$\begin{aligned} N(P_1' P_2' \ldots P_n') &= N - \left|A_1 \cup A_2 \cup \cdots \cup A_n\right| \\ &= N - \sum_{1 \le i \le n} N(P_i) + \sum_{1 \le i < j \le n} N(P_i P_j) - \sum_{1 \le i < j < k \le n} N(P_i P_j P_k) + \\ &\quad \cdots + (-1)^n N(P_1 P_2 \ldots P_n). \end{aligned}$$

Alternative form

Find the number of non-negative integer solutions of

$$x_1 + x_2 + x_3 = 11$$

such that $x_1 \leq 3, x_2 \leq 4$ *and* $x_3 \leq 6$.

Define the properties as follows:

- P_1: $x_1 > 3$,
- P_2: $x_2 > 4$,
- P_3: $x_3 > 6$.

Then the number of non-negative integer solutions such that $x_1 \leq 3, x_2 \leq 4$ and $x_3 \leq 6$ is $N(P_1' P_2' P_3')$.

We have:
$N = C(3 + 11 - 1, 11) = 78$,
$N(P_1) = C(3 + 7 - 1, 7) = 36$, $N(P_2) = C(3 + 6 - 1, 6) = 28$, $N(P_3) = C(3 + 4 - 1, 4) = 15$,
$N(P_1 P_2) = C(3 + 2 - 1, 2) = 6$, $N(P_1 P_3) = C(3 + 0 - 1, 0) = 1$, $N(P_2 P_3) = 0$,
$N(P_1 P_2 P_3) = 0$.

Hence $N(P_1' P_2' P_3') = 78 - 36 - 28 - 15 + 6 + 1 + 0 - 0 = 6$.

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The number of surjective functions

How many surjective functions are there from $\{1,2,3,4,5,6\}$ *to* $\{1,2,3\}$*?*

Let P_i $(i = 1,2,3)$ be the property that i is *not* in the range of the function. Then a function is surjective iff. it does not have any of the properties P_1, P_2, P_3.

By the product rule:

$$\begin{aligned} N &= 3^6 = 729 \\ N(P_i) &= 2^6 = 64 \\ N(P_i P_j) &= 1^6 = 1 \\ N(P_1 P_2 P_3) &= 0^6 = 0\,. \end{aligned}$$

Hence $N(P_1' P_2' P_3') = 729 - 64 - 64 - 64 + 1 + 1 + 1 - 0 = 540$.

The number of surjective functions

Theorem

The number of surjective functions from a set of m elements to a set of n elements, with $m \geq n$ is

$$n^m - C(n,1)(n-1)^m + C(n,2)(n-2)^m - \cdots + (-1)^{n-1} C(n,n-1)1^m .$$

How many ways are there to assign five different jobs to four different employees if every employee is assigned at least one job?

There are $4^5 - C(4,1)3^5 + C(4,2)2^5 - C(4,3)1^5 = 1024 - 972 + 192 - 4 = 240$ ways.

The Stirling numbers of the second kind

A surjective function from a set of m elements to a set of n elements corresponds to a way of:

1. putting m different balls into n identical boxes so that none of the boxes is empty $\Longrightarrow$ $S(m, n)$ ways
2. then label the boxes with different labels $\Longrightarrow$ $n!$ ways

By the product rule, $\#\{\text{surjective functions}\} = S(m, n) \times n!$

Theorem

$$\begin{aligned} S(m,n) &= \frac{1}{n!}\left(n^m - C(n,1)(n-1)^m \right. \\ &\quad \left. + C(n,2)(n-2)^m - \cdots + (-1)^{n-1} C(n,n-1)1^m\right) . \end{aligned}$$

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Derangements

Definition

A derangement of n elements is a permutation of those elements which leaves none of the elements in its original position.

The number of derangements of a set of n elements is denoted by D_n.

Examples:

- $(2, 1, 4, 5, 3)$ is a derangement of the set $\{1, 2, 3, 4, 5\}$.
- $(2, 1, 4, 3, \mathbf{5})$ is not a derangement because 5 is a fixed element.
- There are two derangements of size 3: $(2, 3, 1)$ and $(3, 1, 2)$, so $D_3 = 2$.

Derangements

Theorem

$$D_n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}\right) .$$

Proof: Let P_i be the property (of a permutation) that the element i is fixed. Then $D_n = N(P_1' P_2' \ldots P_n')$.

For $m = 1, \ldots, n$, $N(P_{i_1} P_{i_2} \ldots P_{i_m})$ is the number of permutations that fix the elements $i_1, i_2, \ldots, i_m$. The $(n - m)$ remaining elements are "free", so $N(P_{i_1} P_{i_2} \ldots P_{i_m}) = (n - m)!$.

By the inclusion–exclusion principle:

$$D_n = n! - \sum_{m=1}^{n} (-1)^{m-1} C(n, m)(n - m)! = n! - \sum_{m=1}^{n} (-1)^{m-1} \frac{n!}{m!} ,$$

or

$$D_n = n! \sum_{m=0}^{n} (-1)^m \frac{1}{m!} .$$

Derangements

The Hatcheck Problem
A new employee checks the hats of n people at a restaurant, forgetting to put claim check numbers on the hats. When customers return for their hats, the checker gives them back hats chosen at random from the remaining hats. What is the probability that no one receives the correct hat?

The probability is

$$\frac{D_n}{n!} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}\,.$$

When $n \to \infty$, the RHS converges to $e^{-1} \approx 37\%$.